

AJAY KRISHNA DEVULAPALLY

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PROFESSIONAL SUMMARY

Software engineer and M.S. Computer Science graduate from Penn State with experience building Python and C/C++ systems, backend services, automated evaluation pipelines, and performance-focused software. Strong foundations in data structures, algorithms, operating systems, databases, object-oriented design, REST APIs, debugging, testing, Linux, Git, Docker, and modular software development.

EDUCATION

The Pennsylvania State University

M.S. in Computer Science, Thesis Track

University Park, PA

Aug 2024 – Aug 2026

CVR College of Engineering

B.Tech. in Computer Science and Engineering

Hyderabad, India

2019 – 2023

EXPERIENCE

Graduate Research Assistant

Aug 2024 – Aug 2026

Microsystems Design Lab, Penn State | SRC

University Park, PA

- Conducted SRC-supported research under **Prof. Vijaykrishnan Narayanan** on software systems for AI/ML-driven optimization, chiplet placement, and combinatorial design problems.
- Built modular Python and C++ workflows integrating simulation, optimization, data processing, batch execution, configuration management, and automated benchmark evaluation.
- Implemented and validated GNN-assisted freeze-mask optimization pipelines for oscillator-based Ising machines solving Max-Cut, using graph features, stability analysis, and dynamic node-freezing.
- Reduced OIM runtime by 33.8% on average across 7 external MQLib graphs versus the no-freeze baseline, with up to 54.1% runtime saved and 0.122% median Max-Cut loss.

AI Intern

Mar 2021 – Apr 2021

Smartknower

Bangalore, India

- Achieved 98.3% accuracy on MNIST by building ANN and transfer-learning image-classification models using TensorFlow and Keras.
- Implemented preprocessing, validation, evaluation, error analysis, and Streamlit demonstration workflows using Python, OpenCV, and Pandas.

RESEARCH

TWIRLeT-2.5D – First Author

2026

[Submitted to *TVLSI*] *Ajaykrishna Devulapally, Varun Parekh, Anusha Devulapally, Mirco Sciuilli, Srivatsa Rangchar Srinivasa, Vijaykrishnan Narayanan, “TWIRLeT-2.5D: Thermal Wirelength Integrated Reinforcement Learning for 2.5D Chiplet Placement.”*

- Built an end-to-end Python/PyTorch system integrating action validation, simulation, routing, thermal analysis, configuration handling, and automated benchmark execution.
- Reduced wirelength by up to 27% and achieved 58× faster time-to-solution through reusable inference and evaluation workflows.

PROJECTS

Experiment Registry Backend Service | FastAPI, PostgreSQL, Docker

2026 – In Progress

- Building a REST service for storing experiment metadata, parameters, metrics, and artifacts using FastAPI, SQLAlchemy, PostgreSQL, request validation, and structured error handling.
- Adding Pytest unit and integration tests, Docker Compose for local deployment, and GitHub Actions for automated testing on each push.

Preemptive Thread Scheduler | C, Linux, Operating Systems

2025

- Built a tick-accurate scheduler implementing FCFS, SRTF, and five-level MLFQ with preemption, I/O blocking, priority boosting, deterministic state transitions, and queue management.
- Validated behavior against expected Gantt traces and debugged edge cases involving preemption, blocking, tie-breaking, and workload transitions.

Buddy and Slab Memory Allocator | C, Operating Systems

2025

- Implemented a custom allocator with chunk splitting, recursive coalescing, free-list management, metadata headers, and deterministic address selection.
- Built `my_malloc` and `my_free` with tests for allocation, deallocation, merging, object reuse, and fragmentation behavior.

SKILLS

Programming: Python, C, C++, Java, SQL, JavaScript, TypeScript

Backend / Databases: FastAPI, REST APIs, PostgreSQL, SQLAlchemy, relational databases, request validation, structured logging

Software Engineering: Object-oriented programming, modular design, data structures, algorithms, debugging, Pytest, unit and integration testing

Systems / Tools: Linux, multithreading, synchronization, memory management, Git, GitHub, Docker, GitHub Actions

AI / ML: PyTorch, TensorFlow, Scikit-learn, model evaluation and experimentation

Certificates: Artificial Intelligence – Smartknower (2021), Database Programming with SQL – Oracle (2022)